Koine — Recommended minimal vocabulary set for interoperable knowledge graphs (general purpose)

Evidence-based update of the 2025 ‘Semantic Model Standards’ guide

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Koine — general purpose

Koine (from the common Greek of the Hellenistic world) is a minimal set of widely adopted vocabularies, plus one canonical pattern per modelling need and a query contract. With it, independent people and AI agents can write knowledge graphs that others can query without coordinating in advance. This document gives the general-purpose set. A companion document covers biology and health. The machine-readable form is the `koine` agent skill (`SKILL.md`), which AI agents load when they model or query data.

1. The recommendation

Layer	Vocabulary	Scope in Koine
Default	schema.org (`http://schema.org/`)	everything that schema.org can express: entities, names, identifiers, roles, events, places, organisations, creative works, datasets, offers, actions, quantities
Core	SKOS	concept schemes, taxonomies, code lists, mappings
Core	PROV-O (8 terms)	provenance: `wasDerivedFrom`, `wasGeneratedBy`, `used`, `wasAssociatedWith`, `wasAttributedTo`, `wasRevisionOf`, `Activity`, `SoftwareAgent`
Core	SOSA (14 terms)	measurements and study results: observation, feature of interest, observed property, result, result time, sensor, procedure, sample
Core	DCMI Terms (5 terms)	gaps of schema.org: `replaces`, `conformsTo`, `accrualPeriodicity`, `dateSubmitted`, `dateAccepted`
Base	RDF, RDFS, OWL, XSD	statements (`rdf:Statement`), labels, negation (`owl:NegativePropertyAssertion`), deprecation, datatypes
Module	ODRL 2.2	licences and policies with permissions, prohibitions, duties
Module	CiTO (6 terms)	why a work cites another

Layer	Vocabulary	Scope in Koine
Value spaces	Wikidata IRIs, ISO 3166-1, ISO 4217, BCP 47, IANA media types, UCUM, Creative Commons/SPDX, EU authority tables	values, not vocabularies

The vocabularies are only half of Koine. The other half is the contract:

- one canonical pattern per modelling need: role with a period, dated happening, measurement, third-party claim, negation, absence, qualifier, open key;
- a fixed set of tokens: certainty, absence values, fixed keys;
- query paths that consumers use without seeing the data.

The contract made the difference in our experiments (Section 3).

2. How to use it

- AI agents: install the skill (copy `koine/` to `~/.claude/skills/` or a project's `.claude/skills/`); it is loaded automatically when an agent converts information into RDF or queries a graph.
- People: read `SKILL.md` §2 (rules), §3 (decision table), §5 (query contract) and §6 (patterns). The essentials:
  1. Every entity is an IRI with one type and a `schema:name`. Every code-like designation is also a `schema:identifier`.
  2. Use a schema.org property when one exists; otherwise use an open key (`schema:additionalProperty` with `schema:propertyID` and `schema:value`). Never invent terms in standard namespaces.
  3. Never drop a qualifier (status, scope, rank, comparison group), and never leave a number, date or bound only in text.
  4. Use roles for qualified links, actions for dated happenings, SOSA observations for measurements, `rdf:Statement` for attributed or hedged claims, `schema:Quotation` for opinions, and `owl:NegativePropertyAssertion` for negation.
  5. Never guess a code; label every external code with its official label.

3. Why this set: the evidence

The recommendation comes from a study in which language-model agents encoded documents with each candidate guidance, blinded agents rebuilt the documents from the graphs alone, and consumer agents queried graphs they had never seen (full method and results: accompanying article).

Evidence	Result
Adoption (Web Data Commons 2024, Web Almanac, LOV, GitHub)	structured data, mostly schema.org, on 44% of web domains; DCMI Terms most reused in Linked Data; SKOS, PROV-O, SOSA widely used in their communities
AI-readiness (decode/encode probes, 3 models)	schema.org 0.96, SKOS 1.00, PROV-O 1.00, SOSA 1.00, DCMI Terms 0.96, ODRL 0.94, CiTO 0.82; opaque-ID ontologies (RO 0.30, SIO 0.04) and Wikidata properties (0.21) low
Expressivity (56 general modelling needs, 3 experts, $\alpha = 0.90$)	schema.org alone covers 83% of needs; SKOS, PROV-O and SOSA are the best additions when weighted by adoption and readiness
Development (6 versions, 10 documents)	strict blind-query success 0.46 (2025 guide) → 0.89 (Koine V4); invented terms 13% → 0.4%; term agreement between modelers 0.70 → 0.96; fidelity non-inferior

Evidence	Result
Held-out evaluation (6 general documents + 2 real records)	strict blind-query success 0.63 with Koine vs 0.38 with the 2025 guide and 0.19 without guidance (+24.9 points, 95% CI 18.4–31.7); real records 0.75 vs 0.28; recall non-inferior; invented terms 0.8% vs 9.2%

4. What changed from the 2025 guide

- schema.org is the default for everything it covers. DCMI Terms are reduced to five terms that fill schema.org's gaps.
- PROV-O and SOSA are added for provenance and measurements, two needs the 2025 set covered poorly.
- CiTO becomes an optional module and ODRL is added as an optional module.
- Non-existent terms are removed. 11 of the 113 terms recommended in 2025 do not exist (e.g. `cito:Claim`, `dcterms:Text`). Every Koine term is checked against the official vocabulary files.
- A pattern contract is added. It accounts for most of the gain in interoperability between agents.

5. Considered and not recommended

Candidate	Reason
DCAT	a competing dataset path next to `schema:Dataset`; use the published DCAT-AP ↔ schema.org mapping when exchanging with DCAT catalogues
FOAF, Dublin Core elements, Open Graph	superseded by schema.org for the same needs
Wikidata properties	high coverage but the lowest AI-readiness of all candidates (0.21); use Wikidata item IRIs as values via `schema:sameAs`
OWL-Time, GeoSPARQL, QUDT	heavier than needed: typed XSD dates, `schema:GeoCoordinates`/`GeoShape` and UCUM unit codes cover the measured needs
RO, SIO, IAO (opaque numeric identifiers)	low AI-readiness (0.04–0.30): models do not recall their terms
`sosa:phenomenonTime`, `prov:startedAtTime`	replaced by `sosa:resultTime` and `schema:startTime` / `schema:endTime` for one time slot per fact

6. Limitations

The study used Claude-family models only. The benchmark is synthetic apart from four public records. Absolute query success on unseen domains is lower than on development documents. Treat Koine as a tested starting point: extend it through open keys and modules rather than through new vocabularies, and test extensions with the same encode–decode–query protocol.